sopvar: Mathematical Description and Matlab Implementation

Sachin Shivakumar[1]

[1] *Dr. Shivakumar is with Center for Nonlinear Studies and Applied Mathematics & Plasma Physics Division of Los Alamos National Laboratory, Los Alamos, NM, 87544. {Email: sshivakumar@lanl.gov, sachin.shivakumar@asu.edu}*

August 10, 2026

1 Introduction

In this document, we will develop the PI operators mapping between mixed L_2 spaces. More specifically, we consider the class of operators with domain and range

$$\mathcal{P} : L_2^p[s_i, \cdots, s_m] \to L_2[s_j, \cdots, s_n].$$

With loss of generality, we assume $s_i \in [0, 1]$. Note that the index i to m may not include all integers in the set $\{i, \cdots, m\}$. Likewise for the indices in the range.

We will represent this as a series of operators on 1D space composed with each other, i.e.,

$$\mathcal{P}\mathbf{x}(s_j, \cdots, s_n) = \underbrace{\Gamma_{s_j} \circ \cdots \circ \Gamma_{s_n}}_{\text{Output spatial variables}} \circ \underbrace{\Gamma_{s_i} \circ \cdots \circ \Gamma_{s_m}}_{\text{Input spatial variables}} \mathbf{x}(s_i, \cdots, s_m),$$

where each Γ_{s_k} satisfies the following:

- if s_k is in both domain and range spaces, then Γ_{s_k} is a 3-PI operator with 3 parameters.
- if s_k is only in domain, then Γ_{s_k} is an integral operator $\int_0^1 (\cdot) ds_k$ with a single parameter.
- if s_k is only in the range, then Γ_{s_k} is a multiplier operator $M(s_k, \cdots)\mathbf{x}(\cdots)$, with a single parameter.

So, let us define the indicator function $\mathbf{I}(s_k) = 1$ if $s_k \in \{s_j, \cdots, s_m\} \cap \{s_i, \cdots, s_n\}$ and 0 otherwise. Then the number of parameters are stored in a cell structure whose dimensions are

$$\prod_{k \in \text{union}(\{i,\cdots,m\},\{j,\cdots,n\})} 3^{\mathbf{I}(s_k)}.$$

NOTE, we are not storing Γ_{s_i} individually, but the composed operator. The Γ's serve as a mathematical representation only and to estimate the required number of parameters as well as the structure of the parameters.

Example 1.1. Consider a map from $L_2^p[s_1, s_3, s_4]$ to $L_2^q[s_2, s_3]$. Then,

$$\mathrm{union}(\mathrm{vars_i}, \mathrm{vars_o}) = \{s_1, s_2, s_3, s_4\}, \quad \mathrm{intersection}(\mathrm{vars_i}, \mathrm{vars_o}) = \{s_3\}.$$

All such maps necessarily have the form

$$\begin{aligned}
\mathbf{y}(s_2, s_3) &= \mathcal{P}\mathbf{x}(s_1, s_3, s_4) \\
&= \int_0^1 \int_0^1 N_0(s_2, s_3, \theta_1, \theta_4)\mathbf{x}(\theta_1, s_3, \theta_4)d\theta_1 d\theta_4 \\
&\quad + \int_0^1 \int_0^{s_3} \int_0^1 N_1(s_2, s_3, \theta_1, \theta_3, \theta_4)\mathbf{x}(\theta_1, \theta_3, \theta_4)d\theta_1 d\theta_3 d\theta_4 \\
&\quad\quad + \int_0^1 \int_{s_3}^1 \int_0^1 N_2(s_2, s_3, \theta_1, \theta_3, \theta_4)\mathbf{x}(\theta_1, \theta_3, \theta_4)d\theta_1 d\theta_3 d\theta_4 \\
&= \left(\int_0^1 M_1(\theta_1, \cdots)(\cdot)\right) \circ (M_2(s_2, \cdots)(\cdot)) \\
&\quad \circ \left(M_{30}(s_3, \cdots)(\cdot) + \int_0^{s_3} M_{31}(s_3, \theta_3, \cdots)(\cdot) + \int_{s_3}^1 M_{32}(s_3, \theta_3, \cdots)(\cdot)\right) \\
&\quad \circ \left(\int_0^1 M_4(\theta_4, \cdots)(\cdot)\right) \mathbf{x}(s_1, s_3, s_4).
\end{aligned}$$

Again, note that we do not compute or store M_i's since the factorization of N_i's is difficult and unnecessary.

Then, we need one integral to integrate out s_1, one multiplier to introduce s_2, one 3-PI operators for s_3, and one integral to integrate out s_4. Thus, we have a cell structure for the parameters given by $1 \times 1 \times 3 \times 1$ corresponding to the spatial variables $\{s_1, s_2, s_3, s_4\}$. Each element in this cell structure is a polynomial matrix of the size $q \times p$.

In this example, we store

```
params{1,1,1,1} = N0,  params{1,1,2,1} = N1,   params{1,1,3,1} = N2.
```

2 Addition

Clearly, for addition to be well-defined, we need the matrix dimensions, cell dimensions, $\mathrm{vars_i}$ and $\mathrm{vars_o}$ to match.

Algorithm: $A + B$. Trivial. For each element in cell-structure of A, add corresponding element from B.

3 Adjoint

Going back to the example, the adjoint mapping $L_2[s_2, s_3]$ to $L_2[s_1, s_3, s_4]$. Multiplier in s_2 turns to a integral, whereas integrals in s_1, s_4 turn to multipliers. Parameters for 3-PI in s_3 have standard adjoint. This gives up the operator

$$\begin{aligned}
\mathcal{P}^*\mathbf{y}(s_2, s_3) &= \int_0^1 N_0(\theta_2, s_3, s_1, s_4)^T \mathbf{x}(\theta_2, s_3)d\theta_2 \\
&\quad + \int_0^1 \int_0^{s_3} N_2(\theta_2, \theta_3, s_1, s_3, s_4)^T \mathbf{x}(\theta_2, \theta_3)d\theta_3 d\theta_2 \\
&\quad\quad + \int_0^1 \int_{s_3}^1 N_1(\theta_2, \theta_3, s_1, s_3, s_4)^T \mathbf{x}(\theta_2, \theta_3)d\theta_3 d\theta_2.
\end{aligned}$$

The cell dimensions of the transpose remains unchanged. Just the parameters are transposed, input and output variables are switched, normal and dummy variables are switched in semisep integrals. In multiplier along s_3, only the normal and dummy variables of non-common variables $\{s_2, s_1, s_4\}$ are swapped. In this example, we store

```
params{1,1,1,1} = var_swap(N0',{s1,s2,s4},{t1,t2,t4}),
params{1,1,2,1} = var_swap(N2',{s1,s2,s3,s4},{t1,t2,t3,t4}),
params{1,1,3,1} = var_swap(N1',{s1,s2,s3,s4},{t1,t2,t3,t4}).
```

Algorithm: A'. First, let the size of the *A.params* cell be of the form

$$l_1 \times l_2 \times \cdots \times l_d,$$

where $l_i \in \{1, 3\}$. For each element in params, N_k, with index $(l_1, \cdots, l_d) \in \{1, 3\}^d$,

- first transpose the matrix, N_k.
- for all indices inside $(l_1, \cdots, l_d)$ that have value 2, replace by index 3. Replace indices 3 by 2. For example, index $(1, 2, 3, 1, 2)$ would be changed to $(1, 3, 2, 1, 3)$.
- for each index l_i in $(l_1, \cdots, l_d)$ if dimension along l_i is 1, then swap (s_i, θ_i) in N_k.
- for each index, if dimension along l_i is 3 and current index $l_i = 1$, then do not swap variables; otherwise swap if $l_i = 2$ or $l_i = 3$. From the previous example, for the parameter located at $(1, 1, 1, 1)$ we have $l_3 = 1$ but dimension along l_3 is 3, so do not swap s_3, θ_3. Next for $(1, 1, 2, 1)$, we have dimension along l_3 is 3, but we still swap s_3, θ_3 since $l_3 = 2$. For all other incides, l_1, l_2, l_4, dimensions along them are 1, so always swap s_i, θ_i for $i = 1, 2, 4$.

4 New Introduction

Partition variables as $s_1 \in \mathbb{R}^{n_1}$, $s_2 \in \mathbb{R}^{n_2}$, $s_3 \in \mathbb{R}^{n_3}$ so that

$$\mathcal{P} : L_2^p[s_1, s_3] \to L_2[s_2, s_3].$$

Operators take the form

$$(\mathcal{P}\mathbf{x}(s_1, s_3))(s_2, s_3) = \sum_{\alpha \in \{0,1,-1\}^{n_3}} \int_{s_1} \int_{s_3'} K_\alpha(s_2, s_3, s_3', s_1) x(s_1, s_3') ds_1 ds_3'$$

Where

$$K_\alpha(s_2, s_3, s_3', s_1) = Z_1(s_2, s_3)^T C_\alpha Z_2(s_1, s_3') I_\alpha(s_3 - s_3')$$

where

$$I_\alpha(s) = \prod_i I_{\alpha_i}(s_i) \qquad I_{\alpha_i}(s_i) = \begin{cases} \delta(s_i) & \alpha_i = 0 \\ I(s_i) & \alpha_i = 1 \\ I(-s_i) & \alpha_i = -1 \end{cases} \qquad I(s) = \begin{cases} 1 & s \geq 0 \\ 0 & s < 0 \end{cases}$$

5 New Adjoint

The adjoint property is $\langle y, Ax\rangle = \langle A^* y, x\rangle = \langle x, A^* y\rangle$. Hence we require

$$\begin{aligned}
&\int_{s_2,s_3} y(s_2,s_3)^T \int_{s_3',s_1} K(s_2,s_3,s_3',s_1)x(s_1,s_3')ds_i \\
&= \int_{s_2,s_3}\int_{s_3',s_1} x(s_1,s_3')^T K(s_2,s_3,s_3',s_1)^T y(s_2,s_3)ds_i \\
&= \int_{s_3',s_1} x(s_1,s_3')^T \int_{s_2,s_3} K(s_2,s_3,s_3',s_1)^T y(s_2,s_3)ds_i \\
&= \int_{s_3,s_1} x(s_1,s_3)^T \int_{s_2,s_3'} K(s_2,s_3',s_3,s_1)^T y(s_2,s_3')ds_i
\end{aligned}$$

Now, since $K = \sum_\alpha K_\alpha$, we have

$$K(s_2,s_3',s_3,s_1)^T = \sum_\alpha K_\alpha(s_2,s_3',s_3,s_1)^T$$

Now, since each K_α has the form

$$K_\alpha(s_2,s_3,s_3',s_1) = Z_1(s_2,s_3)^T C_\alpha Z_2(s_1,s_3')I_\alpha(s_3-s_3')$$

$$K_\alpha(s_2,s_3',s_3,s_1) = Z_1(s_2,s_3')^T C_\alpha Z_2(s_1,s_3)I_\alpha(s_3'-s_3)$$

we have

$$K_\alpha(s_2,s_3',s_3,s_1)^T = (Z_1(s_2,s_3')^T C_\alpha Z_2(s_1,s_3))^T I_\alpha(s_3'-s_3) = Z_2(s_1,s_3)^T C_\alpha^T Z_1(s_2,s_3')I_\alpha(s_3'-s_3).$$

Finally, since

$$I_\alpha(s) = \prod_i I_{\alpha_i}(s_i) \qquad I_{\alpha_i}(s_i) = \begin{cases}\delta(s_i) & \alpha_i = 0\\ I(s_i) & \alpha_i = 1\\ I(-s_i) & \alpha_i = -1\end{cases}$$

we have

$$I_\alpha(-s) = \prod_i I_{\alpha_i}(-s_i) \qquad I_{\alpha_i}(-s_i) = \begin{cases}\delta(s_i) & \alpha_i = 0\\ I(s_i) & \alpha_i = -1\\ I(-s_i) & \alpha_i = 1\end{cases}$$

So that $I_\alpha(-s) = I_{-\alpha}(s)$. This implies

$$(\mathcal{P}^*\mathbf{x}(s_2,s_3))(s_1,s_3) = \sum_{\alpha\in\{0,1,-1\}^{n_3}} \int_{s_1}\int_{s_3'} K_\alpha^*(s_1,s_3,s_3',s_2)x(s_2,s_3')ds_2 ds_3'$$

$$K_\alpha^*(s_2,s_3',s_3,s_1) = Z_2(s_1,s_3)^T C_{-\alpha}^T Z_1(s_2,s_3')I_\alpha(s_3-s_3').$$

6 New Composition

Find $C = AB$, where we presume image of B is the range of A.

Partition variables in B as $s_{1b}\in\mathbb{R}^{n_{1b}}$, $s_{2b}\in\mathbb{R}^{n_{2b}}$, $s_{3b}\in\mathbb{R}^{n_3}$ so that

$$\mathcal{B} : L_2^p[s_1|s_{3a},s_{3b}] \to L_2[s_{3a},s_{3b}|s_{2a},s_{2b}].$$

and

$$\mathcal{A} : L_2^p[s_{2b},s_{3b}|s_{2a},s_{3a}] \to L_2[s_{2a},s_{3a}|s_4].$$

Our goal is to find the output operator $C = AB$ which will map

$$\mathcal{C} : L_2^p[s_1, s_{3b}; s_{3a}] \to L_2[s_{3a}|s_{2a}, s_4].$$

and have the form

$$(\mathcal{C}\mathbf{x}(s_1|s_{3a}, s_{3b}))(s_{3a}|s_{2a}, s_4) = \sum_{\gamma}^{\{0,1,-1\}^{n3a}} \int_{s_1, s_{3b}, s'_{3a}} K_\gamma^C(s_4, s_{2a}|s_{3a}, s'_{3a}|s_{3b}, s_1)x(s'_{3a}|s_{3b}, s_1)d(s_1, s'_{3a}, s_{3b})$$

where K_γ^C has the form

$$K_\gamma^C = Z(s_4, s_{2a}|s_{3a})^T C_\gamma^C Z(s'_{3a}|s_{3b}, s_1)I_\gamma(s_{3a} - s'_{3a})$$

Operator $\mathcal{B}$ takes the form

$$\begin{aligned}&(\mathcal{B}\mathbf{x}(s_1|s_{3a}, s_{3b}))(s_{3a}, s_{3b}|s_{2a}, s_{2b})\\ &= \sum_{\alpha_a, \alpha_b}^{\{0,1,-1\}^{n3a+n3b}} \int_{s_1, s'_{3a}, s'_{3b}} K^B_{\alpha_a, \alpha_b}(s_{2a}, s_{2b}|s_{3a}, s_{3b}, s'_{3a}, s'_{3b}|s_1)x(s'_{3a}, s'_{3b}|s_1)d(s_1, s'_{3a}, s'_{3b})\end{aligned}$$

where $K^B_{\alpha_a, \alpha_b}$ has the form

$$K^B_{\alpha_a, \alpha_b} = Z(s_{2a}, s_{2b}|s_{3a}, s_{3b})^T C^B_{\alpha_a, \alpha_b} Z(s'_{3a}, s'_{3b}|s_1)I_{\alpha_a}(s_{3a} - s'_{3a})I_{\alpha_b}(s_{3b} - s'_{3b})$$

and operator $\mathcal{A}$ takes the form

$$\begin{aligned}&(\mathcal{A}\mathbf{y}(s_{2a}, s_{3a}|s_{2b}, s_{3b}))(s_{2a}, s_{3a}|s_4)\\ &= \sum_{\beta_a, \beta_b}^{\{0,1,-1\}^{n2a+n3a}} \int_{s_{2b}, s_{3b}, s'_{3a}, s'_{2a}} K^A_{\beta_a, \beta_b}(s_4|s_{2a}, s_{3a}, s'_{2a}, s'_{3a}|s_{2b}, s_{3b})y(s'_{2a}, s'_{3a}|s_{2b}, s_{3b})d(s'_{2a}, s'_{3a}, s_{2b}, s_{3b})\end{aligned}$$

where $K^A_{\beta_a, \beta_b}$ has the form

$$K^A_{\beta_a, \beta_b} = Z(s_4|s_{2a}, s_{3a})^T C^A_{\beta_a, \beta_b} Z(s'_{2a}, s'_{3a}|s_{2b}, s_{3b})I_{\beta_a}(s_{2a} - s'_{2a})I_{\beta_b}(s_{3a} - s'_{3a})$$

Now, substituting $y = Bx$, we get

$$y(s'_{2a}, s'_{3a}|s_{2b}, s_{3b}) = \sum_{\alpha_a, \alpha_b}^{\{0,1,-1\}^{n3a+n3b}} \int_{s_1, s''_{3a}, s''_{3b}} K^B_{\alpha_a, \alpha_b}(s'_{2a}, s_{2b}|s'_{3a}, s_{3b}, s''_{3a}, s''_{3b}|s_1)x(s''_{3a}, s''_{3b}|s_1)d(s''_{3a}, s''_{3b}|s_1)$$

Thus $\mathcal{A}y = \mathcal{A}\mathcal{B}x$ is

$$
\begin{aligned}
&(\boldsymbol{\mathcal{A}}\mathbf{y}(s_{2a}, s_{3a}|_{2b}, s_{3b}))(s_{2a}, s_{3a}|s_4) \\
&= \sum_{\beta_a,\beta_b}^{\{0,1,-1\}^{n_{2a}+n_{3a}}} \int_{s_{2b},s_{3b},s'_{3a},s'_{2a}} K^A_{\beta_a,\beta_b}(s_4|s_{2a}, s_{3a}, s'_{2a}, s'_{3a}|s_{2b}, s_{3b}) \\
&\qquad \sum_{\alpha_a,\alpha_b}^{\{0,1,-1\}^{n_{3a}+n_{3b}}} \int_{s_1,s''_{3a},s''_{3b}} K^B_{\alpha_a,\alpha_b}(s'_{2a}, s_{2b}|s'_{3a}, s_{3b}, s''_{3a}, s''_{3b}|s_1)x(s''_{3a}, s''_{3b}|s_1)d(s_1, s''_{3a}, s''_{3b}, s'_{2a}, s'_{3a}, s_{2b}, s_{3b}) \\
&= \int_{s_{2b},s_{3b},s'_{3a},s'_{2a},s_1,s''_{3a},s''_{3b}} \sum_{\beta_a,\beta_b,\alpha_a,\alpha_b}^{\{0,1,-1\}^{n_{2a}+n_{3a}+n_{3a}+n_{3b}}} \\
&K^A_{\beta_a,\beta_b}(s_4|s_{2a}, s_{3a}, s'_{2a}, s'_{3a}|s_{2b}, s_{3b})K^B_{\alpha_a,\alpha_b}(s'_{2a}, s_{2b}|s'_{3a}, s_{3b}, s''_{3a}, s''_{3b}|s_1)x(s''_{3a}, s''_{3b}|s_1)d(s_1, s''_{3a}, s''_{3b}, s'_{2a}, s'_{3a}, s_{2b}, s_{3b}) \\
&= \int_{s_1,s''_{3b},s''_{3a}} \sum_{\beta_a,\beta_b,\alpha_a,\alpha_b}^{\{0,1,-1\}^{n_{2a}+n_{3a}+n_{3a}+n_{3b}}} \\
&\left(\int_{s_{2b},s_{3b},s'_{3a},s'_{2a}} K^A_{\beta_a,\beta_b}(s_4|s_{2a}, s_{3a}, s'_{2a}, s'_{3a}|s_{2b}, s_{3b})K^B_{\alpha_a,\alpha_b}(s'_{2a}, s_{2b}|s'_{3a}, s_{3b}, s''_{3a}, s''_{3b}|s_1)d(s'_{2a}, s_{2b}, s'_{3a}, s_{3b})\right) \\
&\qquad x(s''_{3a}, s''_{3b}|s_1)d(s''_{3a}, s''_{3b}, s_1) \\
&= \int_{s_1,s_{3b},s'_{3a}} \sum_{\beta_a,\beta_b,\alpha_a,\alpha_b}^{\{0,1,-1\}^{n_{2a}+n_{3a}+n_{3a}+n_{3b}}} \\
&\left(\int_{s_{2b},s'_{3b},s''_{3a},s'_{2a}} K^A_{\beta_a,\beta_b}(s_4|s_{2a}, s_{3a}, s'_{2a}, s''_{3a}|s_{2b}, s'_{3b})K^B_{\alpha_a,\alpha_b}(s'_{2a}, s_{2b}|s''_{3a}, s'_{3b}, s'_{3a}, s_{3b}|s_1)d(s'_{2a}, s_{2b}, s''_{3a}, s'_{3b})\right) \\
&\qquad x(s'_{3a}, s_{3b}|s_1)d(s'_{3a}, s_{3b}, s_1)
\end{aligned}
$$

So that

$$
(\boldsymbol{\mathcal{A}\mathcal{B}}\mathbf{x}(s_1|s_{3a}, s_{3b}))(s_{3a}|s_{2a}, s_4) = \int_{s_1,s_{3b},s'_{3a}} K^X(s_4, s_{2a}|s_{3a}, s'_{3a}|s_{3b}, s_1)x(s'_{3a}|s_{3b}, s_1)d(s_1, s'_{3a}, s_{3b})
$$

where

$$
\begin{aligned}
&K^X(s_4, s_{2a}|s_{3a}, s'_{3a}|s_{3b}, s_1) = \sum_{\beta_a,\beta_b,\alpha_a,\alpha_b}^{\{0,1,-1\}^{n_{2a}+n_{3a}+n_{3a}+n_{3b}}} K^X_{\beta_a,\beta_b,\alpha_a,\alpha_b}(s_4, s_{2a}|s_{3a}, s'_{3a}|s_{3b}, s_1) \\
&K^X_{\beta_a,\beta_b,\alpha_a,\alpha_b}(s_4, s_{2a}|s_{3a}, s'_{3a}|s_{3b}, s_1) \\
&= \int_{s_{2b},s'_{3b},s''_{3a},s'_{2a}} K^A_{\beta_a,\beta_b}(s_4|s_{2a}, s_{3a}, s'_{2a}, s''_{3a}|s_{2b}, s'_{3b})K^B_{\alpha_a,\alpha_b}(s'_{2a}, s_{2b}|s''_{3a}, s'_{3b}, s'_{3a}, s_{3b}|s_1)d(s'_{2a}, s''_{3a}, s_{2b}, s'_{3b})
\end{aligned}
$$

where from

$$
\begin{aligned}
K^A_{\beta_a,\beta_b}(s_4|s_{2a}, s_{3a}, s'_{2a}, s'_{3a}|s_{2b}, s_{3b}) &= Z(s_4|s_{2a}, s_{3a})^T C^A_{\beta_a,\beta_b} Z(s'_{2a}, s'_{3a}|s_{2b}, s_{3b}) I_{\beta_a}(s_{2a} - s'_{2a}) I_{\beta_b}(s_{3a} - s'_{3a}) \\
K^A_{\beta_a,\beta_b}(s_4|s_{2a}, s_{3a}, s'_{2a}, s''_{3a}|s_{2b}, s'_{3b}) &= Z(s_4|s_{2a}, s_{3a})^T C^A_{\beta_a,\beta_b} Z(s'_{2a}, s''_{3a}|s_{2b}, s'_{3b}) I_{\beta_a}(s_{2a} - s'_{2a}) I_{\beta_b}(s_{3a} - s''_{3a})
\end{aligned}
$$

and

$$
\begin{aligned}
K^B_{\alpha_a,\alpha_b}(s_{2a}, s_{2b}|s_{3a}, s_{3b}, s'_{3a}, s'_{3b}|s_1) &= Z(s_{2a}, s_{2b}|s_{3a}, s_{3b})^T C^B_{\alpha_a,\alpha_b} Z(s'_{3a}, s'_{3b}|s_1) I_{\alpha_a}(s_{3a} - s'_{3a}) I_{\alpha_b}(s_{3b} - s'_{3b}) \\
K^B_{\alpha_a,\alpha_b}(s'_{2a}, s_{2b}|s''_{3a}, s'_{3b}, s'_{3a}, s_{3b}|s_1) &= Z(s'_{2a}, s_{2b}|s''_{3a}, s'_{3b})^T C^B_{\alpha_a,\alpha_b} Z(s'_{3a}, s_{3b}|s_1) I_{\alpha_a}(s''_{3a} - s'_{3a}) I_{\alpha_b}(s'_{3b} - s_{3b})
\end{aligned}
$$

we have

$$
\begin{aligned}
&K^X_{\beta_a,\beta_b,\alpha_a,\alpha_b}(s_4, s_{2a}|s_{3a}, s'_{3a}|s_{3b}, s_1)\\
&= \int_{s_{2b},s'_{3b},s''_{3a},s'_{2a}} K^A_{\beta_a,\beta_b}(s_4|s_{2a}, s_{3a}, s'_{2a}, s''_{3a}|s_{2b}, s'_{3b}) K^B_{\alpha_a,\alpha_b}(s'_{2a}, s_{2b}|s''_{3a}, s'_{3b}, s'_{3a}, s_{3b}|s_1) d(s'_{2a}, s''_{3a}, s_{2b}, s'_{3b})\\
&= \int_{s_{2b},s'_{3b},s''_{3a},s'_{2a}} Z(s_4|s_{2a}, s_{3a})^T C^A_{\beta_a,\beta_b} Z(s'_{2a}, s''_{3a}|s_{2b}, s'_{3b}) I_{\beta_a}(s_{2a} - s'_{2a}) I_{\beta_b}(s_{3a} - s''_{3a})\\
&\qquad Z(s'_{2a}, s_{2b}|s''_{3a}, s'_{3b})^T C^B_{\alpha_a,\alpha_b} Z(s'_{3a}, s_{3b}|s_1) I_{\alpha_a}(s''_{3a} - s'_{3a}) I_{\alpha_b}(s'_{3b} - s_{3b}) d(s'_{2a}, s''_{3a}, s_{2b}, s'_{3b})\\
&= Z(s_4|s_{2a}, s_{3a})^T C^A_{\beta_a,\beta_b} K^Z_{\alpha,\beta}(s_{2a}, s_{3a}|s'_{3a}, s_{3b}) C^B_{\alpha_a,\alpha_b} Z(s'_{3a}, s_{3b}|s_1)
\end{aligned}
$$

where

$$
\begin{aligned}
K^Z_{\alpha,\beta}(s_{2a}, s_{3a}|s'_{3a}, s_{3b}) = &\int_{s'_{3b},s''_{3a},s'_{2a}}\\
&\left(\int_{s_{2b}} Z(s'_{2a}, s''_{3a}|s_{2b}, s'_{3b}) Z(s'_{2a}, s_{2b}|s''_{3a}, s'_{3b})^T d(s_{2b})\right)\\
&I_{\beta_a}(s_{2a} - s'_{2a}) I_{\beta_b}(s_{3a} - s''_{3a}) I_{\alpha_a}(s''_{3a} - s'_{3a}) I_{\alpha_b}(s'_{3b} - s_{3b})\\
&d(s'_{2a}, s''_{3a}, s'_{3b})
\end{aligned}
$$

We may re-write

$$
\int_{s_{2b}} Z(s'_{2a}, s''_{3a}|s_{2b}, s'_{3b}) Z(s'_{2a}, s_{2b}|s''_{3a}, s'_{3b})^T d(s_{2b}) = \hat{Z}(s'_{2a}) G(s''_{3a}) \hat{Z}(s'_{3b})^T
$$

So that

$$
\begin{aligned}
&K^Z_{\alpha,\beta}(s_{2a}, s_{3a}|s'_{3a}, s_{3b})\\
&= \int_{s'_{3b},s''_{3a},s'_{2a}} \hat{Z}(s'_{2a}) G(s''_{3a}) \hat{Z}(s'_{3b}) I_{\beta_a}(s_{2a} - s'_{2a}) I_{\beta_b}(s_{3a} - s''_{3a}) I_{\alpha_a}(s''_{3a} - s'_{3a}) I_{\alpha_b}(s'_{3b} - s_{3b}) d(s'_{2a}, s''_{3a}, s'_{3b})\\
&= \left(\int_{s'_{2a}} \hat{Z}(s'_{2a}) I_{\beta_a}(s_{2a} - s'_{2a})\right) \left(\int_{s''_{3a}} G(s''_{3a}) I_{\alpha_a}(s''_{3a} - s'_{3a}) I_{\beta_b}(s_{3a} - s''_{3a})\right) \left(\int_{s'_{3b}} \hat{Z}(s'_{3b})^T I_{\alpha_b}(s'_{3b} - s_{3b})\right)\\
&= K^{\hat{Z}}_{\beta_a}(s_{2a}) K^G_{\alpha_a,\beta_b}(s_{3a}, s'_{3a}) K^{\hat{Z}}_{\alpha_b}(s_{3b})
\end{aligned}
$$

where

$$
K^G_{\alpha_a,\beta_b}(s_{3a}, s'_{3a}) = \int_{s''_{3a}} G(s''_{3a}) I_{\alpha_a}(s''_{3a} - s'_{3a}) I_{\beta_b}(s_{3a} - s''_{3a}) d(s''_{3a})
$$

Summary of Summation

$$
\begin{aligned}
&K^X(s_4, s_{2a}|s_{3a}, s'_{3a}|s_{3b}, s_1) = \sum_{\beta_a,\beta_b,\alpha_a,\alpha_b}^{\{0,1,-1\}^{n_{2a}+n_{3a}+n_{3a}+n_{3b}}} K^X_{\beta_a,\beta_b,\alpha_a,\alpha_b}(s_4, s_{2a}|s_{3a}, s'_{3a}|s_{3b}, s_1)\\
&= Z(s_4|s_{2a}, s_{3a})^T \left(\sum_{\beta_a,\alpha_b}^{\{0,1,-1\}^{n_{2a}+n_{3b}}} \sum_{\beta_b,\alpha_a}^{\{0,1,-1\}^{n_{3a}+n_{3a}}} C^A_{\beta_a,\beta_b} K^{\hat{Z}}_{\beta_a}(s_{2a}) K^G_{\alpha_a,\beta_b}(s_{3a}, s'_{3a}) K^{\hat{Z}}_{\alpha_b}(s_{3b}) C^B_{\alpha_a,\alpha_b}\right) Z(s'_{3a}, s_{3b}|s_1)
\end{aligned}
$$

6.1 Formulae for $K^G_{\alpha_a,\beta_b}(s_{3a}, s'_{3a})$

Each $K^G_{\alpha_a,\beta_b}(s_{3a}, s'_{3a})$ has the form

$$
K^G_{\alpha_a,\beta_b}(s_{3a}, s'_{3a}) = \int_{s''_{3a}} G(s''_{3a}) I_{\alpha_a}(s''_{3a} - s'_{3a}) I_{\beta_b}(s_{3a} - s''_{3a}) d(s''_{3a})
$$

where $G(s''_{3a})$ does not depend on α_a, β_b. So all we need is a analytic formulae for this integral as

$$\int_{s''_{3a}} G(s''_{3a}) I_{\alpha_a}(s''_{3a} - s'_{3a}) I_{\beta_b}(s_{3a} - s''_{3a}) d(s''_{3a}) = \left(\prod_i \int_{s''_{3a,i}} I_{\alpha_{a,i}}(s''_{3a,i} - s'_{3a,i}) I_{\beta_{b,i}}(s_{3a,i} - s''_{3a,i}) d(s''_{3a,i}) \right) G(s''_{3a})$$

Now, note the formula

$$\int_{s''} I_\beta(s - s'') I_\alpha(s'' - s') G(s'')$$

$$= \int_{s''} G(s'') \begin{cases} \delta(s - s'')\delta(s'' - s') = \delta(s - s')\delta(s - s'') & \beta = \alpha = 0 \\ \delta(s - s'')I(s'' - s') = I(s - s')\delta(s - s'') & \beta = 0, \alpha = 1 \\ \delta(s - s'')I(s' - s'') = I(s' - s)\delta(s - s'') & \beta = 0, \alpha = -1 \\ I(s - s'')\delta(s'' - s') = I(s - s')\delta(s'' - s') & \beta = 1, \alpha = 0 \\ I(s - s'')I(s'' - s') = I(s - s')I(s - s'') - I(s - s')I(s' - s'') & \beta = \alpha = 1 \\ I(s - s'')I(s' - s'') = I(s - s')I(s' - s'') + I(s' - s)I(s - s'') & \beta = 1, \alpha = -1 \\ I(s'' - s)\delta(s'' - s') = I(s' - s)\delta(s'' - s') & \beta = -1, \alpha = 0 \\ I(s'' - s)I(s'' - s') = I(s - s')I(s'' - s) + I(s' - s)I(s'' - s') & \beta = -1, \alpha = 1 \\ I(s'' - s)I(s' - s'') = I(s' - s)I(s' - s'') - I(s' - s)I(s - s'') & \beta = \alpha = -1 \end{cases}$$

$$= \begin{cases} I_0(s - s') \int_{s''} G(s'') I_0(s - s'') & \beta = \alpha = 0 \\ I_1(s - s') \int_{s''} G(s'') I_0(s - s'') & \beta = 0, \alpha = 1 \\ I_{-1}(s - s') \int_{s''} G(s'') I_0(s - s'') & \beta = 0, \alpha = -1 \\ I_1(s - s') \int_{s''} G(s'') I_0(s'' - s') & \beta = 1, \alpha = 0 \\ I_1(s - s') \int_{s''} G(s'') I_1(s - s'') - I_1(s - s') \int_{s''} G(s'') I_1(s' - s'') & \beta = \alpha = 1 \\ I_1(s - s') \int_{s''} G(s'') I_1(s' - s'') + I_{-1}(s - s') \int_{s''} G(s'') I_1(s - s'') & \beta = 1, \alpha = -1 \\ I_{-1}(s - s') \int_{s''} G(s'') I_0(s'' - s') & \beta = -1, \alpha = 0 \\ I_1(s - s') \int_{s''} G(s'') I_{-1}(s - s'') + I_{-1}(s - s') \int_{s''} G(s'') I_{-1}(s' - s'') & \beta = -1, \alpha = 1 \\ I_{-1}(s - s') \int_{s''} G(s'') I_1(s' - s'') - I_{-1}(s - s') \int_{s''} G(s'') I_1(s - s'') & \beta = \alpha = -1 \end{cases}$$

This implies we have

$$\int_{s''_i} I_{\alpha_{a,i}}(s''_i - s'_i) I_{\beta_{b,i}}(s_i - s''_i) d(s''_i) G(s''_i)$$
$$= I_0(s_i - s'_i) \left(\Delta_{p(0,\alpha_i,\beta_i)} G\right)(s_i, s'_i) + I_1(s_i - s'_i) \left(\Delta_{p(1,\alpha_i,\beta_i)} G\right)(s_i, s'_i) + I_{-1}(s_i - s'_i) \left(\Delta_{p(-1,\alpha_i,\beta_i)} G\right)(s_i, s'_i)$$

The possible maps are

$$\Delta_{\delta_i} G = \begin{cases} 0 & \delta_i = 0 \\ \int_{s''_i} G(s'') I_0(s_i - s''_i) ds'' & \delta_i = 1 \\ \int_{s''_i} G(s'') I_0(s'_i - s''_i) ds'' & \delta_i = 2 \\ \int_{s''_i} G(s'') I_1(s_i - s''_i) ds'' & \delta_i = 3 \\ \int_{s''_i} G(s'') I_1(s'_i - s''_i) ds'' & \delta_i = 4 \\ \int_{s''_i} G(s'') I_{-1}(s_i - s''_i) ds'' & \delta_i = 5 \\ \int_{s''_i} G(s'') I_{-1}(s'_i - s''_i) ds'' & \delta_i = 6 \\ \int_{s''_i} G(s'') I_1(s_i - s''_i) - I_1(s'_i - s''_i) ds'' & \delta_i = 7 \\ \int_{s''_i} G(s'') I_{-1}(s'_i - s''_i) - I_{-1}(s_i - s''_i) ds'' & \delta_i = 8 \end{cases}$$

$$p(0, \alpha, \beta) = \begin{cases} 1 & \beta = \alpha = 0 \\ 0 & \text{otherwise.} \end{cases}$$

$$p(1,\alpha,\beta) = \begin{cases} 0 & \beta = \alpha = 0 \\ 1 & \beta = 0, \alpha = 1 \\ 0 & \beta = 0, \alpha = -1 \\ 2 & \beta = 1, \alpha = 0 \\ 7 & \beta = \alpha = 1 \\ 4 & \beta = 1, \alpha = -1 \\ 0 & \beta = -1, \alpha = 0 \\ 5 & \beta = -1, \alpha = 1 \\ 0 & \beta = \alpha = -1 \end{cases} \qquad p(-1,\alpha,\beta) = \begin{cases} 0 & \beta = \alpha = 0 \\ 0 & \beta = 0, \alpha = 1 \\ 1 & \beta = 0, \alpha = -1 \\ 0 & \beta = 1, \alpha = 0 \\ 0 & \beta = \alpha = 1 \\ 3 & \beta = 1, \alpha = -1 \\ 2 & \beta = -1, \alpha = 0 \\ 6 & \beta = -1, \alpha = 1 \\ 8 & \beta = \alpha = -1 \end{cases}$$

Applying this map repeatedly yields

$$\int_{s''_{3a}} G(s''_{3a}) I_{\alpha_a}(s''_{3a} - s'_{3a}) I_{\beta_b}(s_{3a} - s''_{3a}) d(s''_{3a}) = \left(\prod_i \int_{s''_{3a,i}} I_{\alpha_{a,i}}(s''_{3a,i} - s'_{3a,i}) I_{\beta_{b,i}}(s_{3a,i} - s''_{3a,i}) d(s''_{3a,i}) \right) G(s''_{3a})$$
$$= \sum_{\gamma}^{\{-1,0,1\}^{n_{3a}}} \left(\Delta_{p(\gamma,\alpha_a,\beta_b)} G \right)(s, s') I_\gamma(s - s')$$

where

$$\Delta_{p(\gamma,\alpha,\beta)} := \prod_i \Delta_{p_i((\gamma_i,\alpha_i,\beta_i))}$$

6.2 Application to the Summation and Conclusion

Recall that we are looking for a $\mathcal{C}$ of the form

$$(\mathcal{C}\mathbf{x}(s_1|s_{3a}, s_{3b}))(s_{3a}|s_{2a}, s_4) = \int_{s_1, s_{3b}, s'_{3a}} K^C(s_4, s_{2a}|s_{3a}, s'_{3a}|s_{3b}, s_1) x(s'_{3a}|s_{3b}, s_1) d(s_1, s'_{3a}, s_{3b})$$

where K^C_γ has the form

$$K^C = Z(s_4, s_{2a}|s_{3a})^T \left(\sum_{\gamma}^{\{0,1,-1\}^{n_{3a}}} C^C_\gamma I_\gamma(s_{3a} - s'_{3a}) \right) Z(s'_{3a}|s_{3b}, s_1)$$

The goal is to construct the constants C^C_γ. Currently, we have $\mathcal{C}$ in the form

$$(\mathcal{C}\mathbf{x}(s_1|s_{3a}, s_{3b}))(s_{3a}|s_{2a}, s_4) = \int_{s_1, s_{3b}, s'_{3a}} K^X(s_4, s_{2a}|s_{3a}, s'_{3a}|s_{3b}, s_1) x(s'_{3a}|s_{3b}, s_1) d(s_1, s'_{3a}, s_{3b})$$

where

$$K^X(s_4, s_{2a}|s_{3a}, s'_{3a}|s_{3b}, s_1)$$
$$= Z(s_4|s_{2a}, s_{3a})^T \left(\sum_{\beta_a, \alpha_b}^{\{0,1,-1\}^{n_{2a}+n_{3b}}} \sum_{\beta_b, \alpha_a}^{\{0,1,-1\}^{n_{3a}+n_{3a}}} C^A_{\beta_a,\beta_b} K^{\hat{Z}}_{\beta_a}(s_{2a}) K^G_{\alpha_a,\beta_b}(s_{3a}, s'_{3a}) K^{\hat{Z}}_{\alpha_b}(s_{3b}) C^B_{\alpha_a,\alpha_b} \right) Z(s'_{3a}, s_{3b}|s_1)$$
$$= Z(s_4|s_{2a}, s_{3a})^T \sum_{\beta_b, \alpha_a}^{\{0,1,-1\}^{2n_{3a}}} \left(\sum_{\beta_a}^{\{0,1,-1\}^{n_{2a}}} C^A_{\beta_a,\beta_b} K^{\hat{Z}}_{\beta_a}(s_{2a}) \right) \left(K^G_{\alpha_a,\beta_b}(s_{3a}, s'_{3a}) \right) \left(\sum_{\alpha_b}^{\{0,1,-1\}^{n_{3b}}} K^{\hat{Z}}_{\alpha_b}(s_{3b}) C^B_{\alpha_a,\alpha_b} \right) Z(s'_{3a},$$

where

$$K^{\hat{Z}}_{\beta_a}(s_{2a}) = \left(\int_{s'_{2a}} \hat{Z}(s'_{2a}) I_{\beta_a}(s_{2a} - s'_{2a}) \right) \qquad K^{\hat{Z}}_{\alpha_b}(s_{3b}) = \left(\int_{s'_{3b}} \hat{Z}(s'_{3b})^T I_{\alpha_b}(s'_{3b} - s_{3b}) \right)$$

where the $\hat{Z}$ are defined using

$$\int_{s_{2b}} Z(s'_{2a}, s''_{3a}|s_{2b}, s'_{3b})Z(s'_{2a}, s_{2b}|s''_{3a}, s'_{3b})^T d(s_{2b}) = \hat{Z}(s'_{2a})G(s''_{3a})\hat{Z}(s'_{3b})^T$$

Before proceeding, let us compute the partial sums

$$L_{\beta_b}(s_{2a}) := \left(\sum_{\beta_a}^{\{0,1,-1\}^{n_{2a}}} C^A_{\beta_a,\beta_b} K^{\hat{Z}}_{\beta_a}(s_{2a})\right) \qquad R_{\alpha_a}(s_{3b}) := \left(\sum_{\alpha_b}^{\{0,1,-1\}^{n_{3b}}} K^{\hat{Z}}_{\alpha_b}(s_{3b}) C^B_{\alpha_a,\alpha_b}\right)$$

and re-arrange terms so that

$$Z(s_4|s_{2a}, s_{3a})^T L_{\beta_b}(s_{2a}) = Z(s_4, s_{2a}|s_{3a})^T C^L_{\beta_b}$$

$$R_{\alpha_a}(s_{3b})Z(s'_{3a}, s_{3b}|s_1) = C^R_{\alpha_a} Z(s'_{3a}|s_{3b}, s_1)$$

We now obtain the simplified form

$$K^X(s_4, s_{2a}|s_{3a}, s'_{3a}|s_{3b}, s_1) = Z(s_4|s_{2a}, s_{3a})^T \left(\sum_{\beta_b,\alpha_a}^{\{0,1,-1\}^{2n_{3a}}} C^L_{\beta_b} K^G_{\alpha_a,\beta_b}(s_{3a}, s'_{3a}) C^R_{\alpha_a}\right) Z(s'_{3a}, s_{3b}|s_1)$$

Now recall

$$K^G_{\alpha_a,\beta_b}(s_{3a}, s'_{3a}) = \left(\int_{s''_{3a}} G(s''_{3a}) I_{\alpha_a}(s''_{3a} - s'_{3a}) I_{\beta_b}(s_{3a} - s''_{3a})\right) = \sum_{\gamma}^{\{-1,0,1\}^{n_{3a}}} \left(\Delta_{p(\gamma,\alpha_a,\beta_b)} G\right)(s_{3a}, s'_{3a}) I_\gamma(s_{3a} - s'_{3a})$$

Rewriting

$$C^L_{\beta_b}\left(\Delta_{p(\gamma,\alpha_a,\beta_b)} G\right)(s_{3a}, s'_{3a}) C^R_{\alpha_a} = \bar{Z}(s_{3a})^T C^G_{\gamma,\beta_b,\alpha_a} \bar{Z}(s'_{3a})$$

We conclude that

$$K^X(s_4, s_{2a}|s_{3a}, s'_{3a}|s_{3b}, s_1) = Z(s_4|s_{2a}, s_{3a})^T \left(\sum_{\gamma}^{\{-1,0,1\}^{n_{3a}}} C^C_\gamma I_\gamma(s_{3a} - s'_{3a})\right) Z(s'_{3a}, s_{3b}|s_1)$$

where

$$C^C_\gamma = C^T_L \left(\sum_{\beta_b,\alpha_a}^{\{0,1,-1\}^{2n_{3a}}} C^G_{\gamma,\beta_b,\alpha_a}\right) C_L$$

and where C_L is such that $\bar{Z}(s'_{3a})Z(s'_{3a}, s_{3b}|s_1) = C_L Z(s'_{3a}, s_{3b}|s_1)$

7 Dsopvar

The obvious Dsopvar representation is basically the same. Partition variables as $s_1 \in \mathbb{R}^{n_1}, s_2 \in \mathbb{R}^{n_2}, s_3 \in \mathbb{R}^{n_3}$ so that

$$\mathcal{P} : L^p_2[s_1, s_3] \to L_2[s_2, s_3].$$

Decision operators take the form

$$(\mathcal{P}\mathbf{x}(s_1, s_3))(s_2, s_3) = \sum_{\alpha \in \{0,1,-1\}^{n_3}} \int_{s_1}\int_{s'_3} K_\alpha(s_2, s_3, s'_3, s_1) x(s_1, s'_3) ds_1 ds'_3$$

Where again

$$K_\alpha(s_2, s'_3, s_3, s_1) = Z_1(s_2, s_3)^T C_\alpha(D) Z_2(s_1, s'_3) I_\alpha(s_3 - s'_3)$$

where

$$I_\alpha(s) = \prod_i I_{\alpha_i}(s_i) \qquad I_{\alpha_i}(s_i) = \begin{cases} \delta(s_i) & \alpha_i = 0 \\ I(s_i) & \alpha_i = 1 \\ I(-s_i) & \alpha_i = -1 \end{cases} \qquad I(s) = \begin{cases} 1 & s \geq 0 \\ 0 & s < 0 \end{cases}$$

and where $C_\alpha(D)$ are matrix-valued affine functions of D where D are the decision variables.

Before focusing on $C(D)$, let consider the the key operations on $C(D)$ required for composition.

Key operation 1: Computing the partial sums $L_{\beta_b}(s_{2a})$ or $R_{\alpha_a}(s_{3b})$

$$L_{\beta_b}(s_{2a}) := \left(\sum_{\beta_a}^{\{0,1,-1\}^{n_{2a}}} C^A_{\beta_a,\beta_b} K^{\hat{Z}}_{\beta_a}(s_{2a}) \right) \qquad R_{\alpha_a}(s_{3b}) := \left(\sum_{\alpha_b}^{\{0,1,-1\}^{n_{3b}}} K^{\hat{Z}}_{\alpha_b}(s_{3b}) C^B_{\alpha_a,\alpha_b} \right)$$

where

$$K^{\hat{Z}}_{\beta_a}(s_{2a}) = \left(\int_{s'_{2a}} \hat{Z}(s'_{2a}) I_{\beta_a}(s_{2a} - s'_{2a}) \right) \qquad K^{\hat{Z}}_{\alpha_b}(s_{3b}) = \left(\int_{s'_{3b}} \hat{Z}(s'_{3b})^T I_{\alpha_b}(s'_{3b} - s_{3b}) \right)$$

Key operation 2: Finding $C^L_{\beta_b}$ or $C^R_{\alpha_a}$ such that

$$Z(s_4|s_{2a}, s_{3a})^T L_{\beta_b}(s_{2a}) = Z(s_4, s_{2a}|s_{3a})^T C^L_{\beta_b}$$

$$R_{\alpha_a}(s_{3b}) Z(s'_{3a}, s_{3b}|s_1) = C^R_{\alpha_a} Z(s'_{3a}|s_{3b}, s_1)$$

We now obtain the simplified form

$$K^X(s_4, s_{2a}|s_{3a}, s'_{3a}|s_{3b}, s_1) = Z(s_4|s_{2a}, s_{3a})^T \left(\sum_{\beta_b,\alpha_a}^{\{0,1,-1\}^{2n_{3a}}} C^L_{\beta_b} K^G_{\alpha_a,\beta_b}(s_{3a}, s'_{3a}) C^R_{\alpha_a} \right) Z(s'_{3a}, s_{3b}|s_1)$$

Now recall

$$K^G_{\alpha_a,\beta_b}(s_{3a}, s'_{3a}) = \left(\int_{s''_{3a}} G(s''_{3a}) I_{\alpha_a}(s''_{3a} - s'_{3a}) I_{\beta_b}(s_{3a} - s''_{3a}) \right) = \sum_{\gamma}^{\{-1,0,1\}^{n_{3a}}} \left(\Delta_{p(\gamma,\alpha_a,\beta_b)} G \right) (s_{3a}, s'_{3a}) I_\gamma(s_{3a} - s'_{3a})$$

Key operation 3: Finding the $C^G_{\gamma,\beta_b,\alpha_a}$ such that

$$C^L_{\beta_b} \left(\Delta_{p(\gamma,\alpha_a,\beta_b)} G \right) (s_{3a}, s'_{3a}) C^R_{\alpha_a} = \bar{Z}(s_{3a})^T C^G_{\gamma,\beta_b,\alpha_a} \bar{Z}(s'_{3a})$$

We conclude that

$$K^X(s_4, s_{2a}|s_{3a}, s'_{3a}|s_{3b}, s_1) = Z(s_4|s_{2a}, s_{3a})^T \left(\sum_{\gamma}^{\{-1,0,1\}^{n_{3a}}} C^C_\gamma I_\gamma(s_{3a} - s'_{3a}) \right) Z(s'_{3a}, s_{3b}|s_1)$$

Key operation 4: Computing the sum C^C_γ

$$C^C_\gamma = C^T_L \left(\sum_{\beta_b,\alpha_a}^{\{0,1,-1\}^{2n_{3a}}} C^G_{\gamma,\beta_b,\alpha_a} \right) C_L$$

and where C_L is such that $\bar{Z}(s'_{3a}) Z(s'_{3a}, s_{3b}|s_1) = C_L Z(s'_{3a}, s_{3b}|s_1)$

7.1 Key Operation 1

$$L_{\beta_b}(s_{2a}) := \left(\sum_{\beta_a}^{\{0,1,-1\}^{n_{2a}}} C^A_{\beta_a,\beta_b} K^{\hat{Z}}_{\beta_a}(s_{2a})\right) \qquad R_{\alpha_a}(s_{3b}) := \left(\sum_{\alpha_b}^{\{0,1,-1\}^{n_{3b}}} K^{\hat{Z}}_{\alpha_b}(s_{3b}) C^B_{\alpha_a,\alpha_b}\right)$$

where

$$K^{\hat{Z}}_{\beta_a}(s_{2a}) = \left(\int_{s'_{2a}} \hat{Z}(s'_{2a}) I_{\beta_a}(s_{2a} - s'_{2a})\right) \qquad K^{\hat{Z}}_{\alpha_b}(s_{3b}) = \left(\int_{s'_{3b}} \hat{Z}(s'_{3b})^T I_{\alpha_b}(s'_{3b} - s_{3b})\right)$$

We may represent $K^{\hat{Z}}_{\beta_a}(s_{2a}) = K_{\beta_a} Z(s_{2a})$ and $K^{\hat{Z}}_{\alpha_b}(s_{3b}) = Z(s_{3b}) K_{\alpha_b}$ using a common basis.

$$L_{\beta_b}(s_{2a}) := \left(\sum_{\beta_a}^{\{0,1,-1\}^{n_{2a}}} C^A_{\beta_a,\beta_b} K_{\beta_a}\right) Z(s_{2a})$$

In this case, we are simply adding decision variables $C^A_{\beta_a,\beta_b} K_{\beta_a}$, which is hopefully not too complex in any build.

7.2 Key Operation 2

The next step is to shift the monomials $Z(s_{2a})$ to the left. Simplifying the problem, we have $C(D)Z(s)$ and we want to construct $\hat{C}(D)$ such that

$$C(D)Z(s) = Z(s)^T \hat{C}(D)$$

However, it seems like this is just a special case of kernel adjoint. That is, if kernel objects have the form $K(s,\theta) = Z_1(s)^T C(D) Z_2(\theta)$, then the adjoint operation on $K(s,\theta)$ is

$$K^*(s,\theta) = K(\theta,s)^T$$

In this case, if we define $\hat{C}(D) = C(D)^T$ and kernel object $K(s,\theta) = \hat{C}(D) Z_2(\theta)$

$$K^*(s,\theta) = K(\theta,s)^T = (\hat{C}(D)Z(s))^T = Z(s)^T \hat{C}(D)^T = Z(s)^T C(D)$$

So the critical operation for Key operation 2 is to have an adjoint operation defined for kernel objects.

7.3 Key Operation 3

Finding the $C^G_{\gamma,\beta_b,\alpha_a}$ such that

$$C^L_{\beta_b}\left(\Delta_{p(\gamma,\alpha_a,\beta_b)} G\right)(s_{3a}, s'_{3a}) C^R_{\alpha_a} = \bar{Z}(s_{3a})^T C^G_{\gamma,\beta_b,\alpha_a} \bar{Z}(s'_{3a})$$

Presuming that $\left(\Delta_{p(\gamma,\alpha_a,\beta_b)} G\right)(s_{3a}, s'_{3a})$ can be represented as

$$\left(\Delta_{p(\gamma,\alpha_a,\beta_b)} G\right)(s_{3a}, s'_{3a}) = Z(s_{3a})^T K^G_{\gamma,\beta_b,\alpha_a} Z(s'_{3a})$$

Then as in key operation 2, we again shift $Z(s_{3a})^T$ to the left and $Z(s'_{3a})$ to the right so that

$$C^L_{\beta_b} Z(s_{3a})^T = Z(s_{3a}) \hat{C}^L_{\beta_b} \qquad Z(s'_{3a}) C^R_{\alpha_a} = \hat{C}^R_{\alpha_a} Z(s'_{3a})$$

In fact, it would be best to combine this shift with the shift in Key operation 2, so that the order is somewhat reversed.

Then all that remains is to multiply

$$\hat{C}^L_{\beta_b} \hat{C}^R_{\alpha_a}$$

Which is a third critical operation on the decision variable structure.

7.4 Key Operation 4

This is pretty straightforward

$$C^C_\gamma = C_L^T \left(\sum_{\beta_b, \alpha_a}^{\{0,1,-1\}^{2n_{3a}}} C^G_{\gamma,\beta_b,\alpha_a} \right) C_L$$

where C_L is fixed. So efficient summation is all that is needed here.

7.5 Conclusion

For efficient composition, we need efficient algorithms for summation of decision variables, kernel adjoint, and multiplication of decision variables.

8 Summation, Adjoint, and Multiplication Operations

Recall our kernel objects take the form

$$K_\alpha(s_2, s_3', s_3, s_1) = Z_1(s_2, s_3)^T C_\alpha(D) Z_2(s_1, s_3') I_\alpha(s_3 - s_3')$$

where we may recall that $Z_2(s_2, s_3)$ is matrix-valued and hence may be represented as $Z_2(s_1, s_3') = Z_d(s_1, s_3') \otimes I_n$ for vector Z_d, which may, in turn, be represented as $Z_d(s_1, s_3') = Z_d(s_3') \otimes Z_d(s_1)$

The key operations on this object which have been identified for composition are summation of the $C_\alpha(D)$, left or right multiplication of $C_\alpha(D)$ by a given matrix, and finding the adjoint of $K_\alpha(s_2, s_3', s_3, s_1)$. To this list, of course, we may add the important step of defining equality constraints between kernels, which requires a common basis. Also, addition of kernels requires a common basis, so this is somewhat important.

So now the big question is how to represent the $C(D)$ which will not result in large computational bottlenecks at these key operations.

Options:

1. The simplest is to make $C_\alpha(D)$ a dpvar so that $C_\alpha(D) = (I \otimes D)^T C$. However, implementation of operations listed below may become tricky

2. Create scalarized "primal" version of dopvar, so that

$$C_\alpha(D) = \sum_i D_i E_i + B$$

   The advantage is that the scalars D_i will exit the integrals and all the previous formulae are applied to the E_i (and B). The disadvantage of this approach is that if there are millions of decision variables, the previous formulae will need to be applied millions of times. The advantage is that the E_i will be relatively small and consist of very few nonzero terms.

3. Create a "dual" version where D is stored as a vector and

$$vecC_\alpha(D) = A + B^T D$$

   of course, the trick here is to translate the operations on $C_\alpha(D)$ to operations on $vecC_\alpha(D)$. B_T is used instead of B in order to minimize the number of columns, which is the major storage cost in the Matlab sparsity structure. Note that this is very similar to the `dpvar` object, but we give up the notation $(I \otimes D)^T C$ and simply think in terms of vec/unvec operations. It may be possible to reuse/repurpose a significant portion of the dpvar code for this object.

4. Create a “LMI” version where
$$C_\alpha(D) = ADC + B$$
where D is a matrix, or for vector representation, the diagonalized version
$$C_\alpha(D) = Adiag(ED)C + B$$
Both these representations have the downside of making addition more complex.

8.1 Key Operations for “dual structure”

The dual structure seems to have a few natural benefits and mirrors to some extent the existing $A^T x = b$ structure used in SOSTOOLS – implying construction of equality constraints may be facilitated using this representation.

8.1.1 Internal Representation

So here we have that $d \in \mathbb{R}^N$ where N is large
$$vecC(d) = A + B^T d$$
Storage is of A, B and metadata, m, n on the dimensions of $C(d)$. We will also, of course, need to store data on the monomial bases, Z_1, Z_2. We may actually want to use a kernel wrapper class here, which includes the monomial bases and the decision variable object (which is stored internally).

The presumption is that $C(d)$ will be defined implicitly, so that if need not be actually constructed in the workspace. Theoretically, however, it can be constructed as needed using the dimensional metadata as
$$C(d) = unvec(A + B^T d, m, n)$$
Note that the unvec operation is linear in d. We may find it useful to store additional data on C in the structure, but will try to avoid doing so.

8.1.2 Multiplication by a constant matrix

This is needed for summation, so lets do it first. kernels are not significant here, so lets focus on the decision variable operation. To begin, we use the identity
$$vec(LCR) = (R^T \otimes L)vec(C)$$
Now if $vec(C(d)) = A + B^T d$, we have
$$vec(LC(d)R) = (R^T \otimes L)A + (R^T \otimes L)B^T d$$
So that the new decision variable object which results from matrix multiplication can be computed through standard matrix multiplication on the elements A, B.

Note: If computing the Kronecker product explicitly is not desired, we may either 1) keep track of L, R in metadata or 2) apply the operation more efficiently by extracting columns of B and applying L^T to each column. I see this as an issue if we do sequential addition operations, and each requires construction of a common monomial basis. Alternatively, we can do a batch sum operation which finds the common basis first and hence avoids repeated transformations.

8.1.3 Summation

Adding two decision variable objects of compatible size is trivial. If $C_1(D) = unvec(A_1 + B_1^T d)$ and $C_2(D) = unvec(A_2 + B_2^T d)$, then

$$C_1(D) + C_1(D) = unvec(A_1 + A_2 + (B_1 + B_2)^T d)$$

Adding two kernel objects requires finding a common monomial basis

$$K_1(s_2, s_3', s_3, s_1) = Z_{a,1}(s_2, s_3)^T C_1(d) Z_{b,1}(s_1, s_3') I_\alpha(s_3 - s_3')$$

$$K_2(s_2, s_3', s_3, s_1) = Z_{a,2}(s_2, s_3)^T C_2(d) Z_{b,2}(s_1, s_3') I_\alpha(s_3 - s_3')$$

Our first step is to find the union of monomials $Z_{a,3} = Z_{a,1} \cup Z_{a,2}$ and $Z_{b,3} = Z_{b,1} \cup Z_{b,2}$ and the associated matrices $T_{a,1}$, $T_{a,2}$,$T_{b,1}$, $T_{b,2}$ such that

$$Z_{a,1} = T_{a,1} Z_{a,3}, \quad Z_{a,2} = T_{a,2} Z_{a,3}, \quad Z_{b,1} = T_{b,1} Z_{b,3}, \quad Z_{b,2} = T_{b,2} Z_{b,3}$$

Then

$$K_1(s_2, s_3', s_3, s_1) = Z_{a,3}(s_2, s_3)^T T_{a,1}^T C_1(d) T_{b,1} Z_{b,3}(s_1, s_3') I_\alpha(s_3 - s_3')$$
$$K_2(s_2, s_3', s_3, s_1) = Z_{a,3}(s_2, s_3)^T T_{a,2}^T C_2(d) T_{b,2} Z_{b,3}(s_1, s_3') I_\alpha(s_3 - s_3')$$

And hence the new kernel object has bases $Z_{a,3}$, $Z_{b,3}$ and decision variable object

$$T_{a,1}^T C_1(d) T_{b,1} + T_{a,2}^T C_2(d) T_{b,2} = unvec(T_1 A_1 + T_1 A_2 + (B_1 T_1^T + B_2 T_1^T)^T d)$$

$$T_1 = T_{b,1}^T \otimes T_{a,1}^T, \qquad T_2 = T_{b,2}^T \otimes T_{a,2}^T$$

Note: This analysis presumes that the list of decision variables in C_1 and C_2 have been synchronized using a combine basis operation.

8.1.4 Combine Basis

Suppose C_1 and C_2 have a common monomial basis and hence have identical dimension

$$C_1(d_1) = unvec(A_1 + B_1^T d_1) \qquad C_2(d_2) = unvec(A_2 + B_2^T d_2)$$

Find the common decision variable basis, d_3, so that $d_1 = T_1 d_3$ and $d_2 = T_2 d_3$

$$C_1(d_1)+C_2(d_2) = unvec(A_1+B_1^T d_1)+nvec(A_2+B_2^T d_2) = unvec(A_1+A_2+B_1^T T_1 d_3+B_2^T T_2 d_3) = unvec(A_1+A_2$$

so the new object is $\{A_1 + A_2, B_1^T T_1 + B_2^T T_2\}$. Note this requires keeping track of decision variable names, as was done in `dpvar`.

8.1.5 Adjoint

$$K_\alpha(s_2, s_3', s_3, s_1) = Z_1(s_2, s_3)^T C_\alpha(d) Z_2(s_1, s_3') I_\alpha(s_3 - s_3')$$

Recall that

$$K_\alpha^*(s_2, s_3', s_3, s_1) = Z_2(s_1, s_3)^T C_{-\alpha}^T Z_1(s_2, s_3') I_\alpha(s_3 - s_3').$$

So that adjoint requires exchange of $C_{-\alpha}(d) = C_\alpha(d)$ and a transpose operation on $C_\alpha(d)$.

The transpose operation requires either a redefinition of the `unvec` operation or for $C(d) = unvec(A + B^T d)$ what I think is simply a re-ordering of the columns of $B' = B(:, p)$ and $A' = A(p)$ for some permutation, p

8.1.6 Construction of Equality Constraints

The primary challenge in construction of equality constraints between kernel objects is again finding a common monomial basis. We won't revisit this problem since it is the same problem as in addition. Presuming we have constructed a common monomial basis, equality constraints are relatively straightforward.

$$K_1(s_2, s_3', s_3, s_1) = Z_{a,3}(s_2, s_3)^T C_1(d) Z_{b,3}(s_1, s_3') I_\alpha(s_3 - s_3')$$

$$K_2(s_2, s_3', s_3, s_1) = Z_{a,3}(s_2, s_3)^T C_2(d) Z_{b,3}(s_1, s_3') I_\alpha(s_3 - s_3')$$

Then $K_1 = K_2$ if and only if $C_1(d) = C_2(d)$ or if $C_1(d) = unvec(A_1 + B_1^T d)$ and $C_2(d) = unvec(A_2 + B_2^T d)$, we require $A_1 = A_2$ and $B_1 = B_2$.

9 Construction of Positive Operators

The basic structure of a positive dsopvar of dimension $m \times m$ is

$$P = \begin{bmatrix} Z_{\alpha_1} \otimes I_m \\ \vdots \\ Z_{\alpha_N} \otimes I_m \end{bmatrix}^* Q \begin{bmatrix} Z_{\alpha_1} \otimes I_m \\ \vdots \\ Z_{\alpha_N} \otimes I_m \end{bmatrix}$$

$$= \sum_{ij} (Z_{\alpha_i} \otimes I_m)^* Q_{ij} (Z_{\alpha_j} \otimes I_m)$$

where the Z_i are sopvar bases to be defined.

Note on `sosquadvar` The workhorse of sostools and dpvar is sosquadvar. This returns a cell array of dpvar objects of the form

$$K_{ij}(s, \theta) = Z_1(s) P_{ij} Z_2(\theta)$$

where you can specify anything for Z_1, Z_2. `sosquadvar` returns this in dpvar format,

$$K_{ij}(s, \theta) = vec(P)^T C Z_{12}(s, \theta)$$

which may actually be more convenient in this case than the new kernel format. If this turns out not to be the case, `sosquadvar` can be modified without too much pain and indeed should be a more efficient translation to the new kernel format. If this is desired (and not yet done or tested), as a temporary measure, we can construct in dpvar and then convert to dsopvar kernel format.

9.1 Basis operators

The structure of the basis requires us to specify the dimensions of the desired positive operator. Of course, the operator must be self-adjoint, so this simplifies things a bit conceptually. Specifically, we can ignore the lost and created variables s_1, s_2 in the general form, which leaves

$$\mathcal{P} : L_2^p[s_3] \to L_2[s_3].$$

Operators take the form

$$(\mathcal{P}\mathbf{x}(s_3))(s_3) = \sum_{\alpha \in \{0,1,-1\}^{n_3}} \int_{s_3'} K_\alpha(s_3, s_3') x(s_3') ds_3'$$

Where

$$K_\alpha(s_3, s_3') = Z_1(s_3)^T C_\alpha Z_2(s_3') I_\alpha(s_3 - s_3')$$

where

$$I_\alpha(s) = \prod_i I_{\alpha_i}(s_i) \qquad I_{\alpha_i}(s_i) = \begin{cases} \delta(s_i) & \alpha_i = 0 \\ I(s_i) & \alpha_i = 1 \\ I(-s_i) & \alpha_i = -1 \end{cases} \qquad I(s) = \begin{cases} 1 & s \geq 0 \\ 0 & s < 0 \end{cases}$$

We presume, then, that each basis function will have this form, but consist of only a single term, so that

$$(\mathcal{Z}_{\alpha_i}\mathbf{x}(s))(s) = I_{\alpha_i}(s-s')\int_{s'} Z(s) \otimes Z(s')x(s')ds'$$

OR

$$(\mathcal{Z}_{\alpha_i}\mathbf{x}(s))(s) = I_{\alpha_i}(s-s')\int_{s'} Z(s,s')x(s')ds'$$

9.2 Construction

Recall we want to construct

$$\mathcal{P} = \sum_{ij}(Z_{\alpha_i} \otimes I_m)^* Q_{ij}(Z_{\alpha_j} \otimes I_m) = \sum_{ij} \mathcal{P}_{ij}$$

where ignoring I_m for now,

$$\mathcal{P}_{ij} = Z^*_{\alpha_i} Q_{ij} Z_{\alpha_j}$$

Recalling the formula for adjoint of

$$(\mathcal{P}\mathbf{x})(s) = \sum_{\alpha \in \{0,1,-1\}^{n_3}} \int_{s'} K_\alpha(s,s')x(s')ds' \qquad K_\alpha(s,s') = Z_1(s)^T C_\alpha Z_2(s') I_\alpha(s-s')$$

is

$$(\mathcal{P}^*\mathbf{x})(s) = \sum_{\alpha \in \{0,1,-1\}^{n_3}} \int_{s'} K^*_\alpha(s,s')x(s')ds' \qquad K^*_\alpha(s',s) = Z_2(s)^T C^T_{-\alpha} Z_1(s') I_\alpha(s-s').$$

Now, noting that a basis operator only has one α, we have that if

$$(\mathcal{Z}_{\alpha_i}\mathbf{x})(s) = I_{\alpha_i}(s-s')\int_{s'} Z(s,s')x(s')ds'$$

then

$$(\mathcal{Z}_{\alpha_i}\mathbf{x})^*(s) = I_{\alpha_i}(s'-s)\int_{s'} Z(s',s)^T x(s')ds'$$

so that

$$\begin{aligned}
&(Z^*_{\alpha_i} Q_{ij} Z_{\alpha_j}\mathbf{x})(s) \\
&= I_{\alpha_i}(s'-s)\int_{s'} Z(s',s)^T Q_{ij} I_{\alpha_j}(s'-s'')\int_{s''} Z(s',s'')x(s'')ds'ds'' \\
&= \int_{s'}\int_{s''} I_{\alpha_i}(s'-s) I_{\alpha_j}(s'-s'') Z(s',s)^T Q_{ij} Z(s',s'')x(s'')ds'ds'' \\
&= \int_{s'}\int_{s''} I_{\alpha_i}(s'-s) I_{\alpha_j}(s'-s'') R_{\alpha_i,\alpha_j}(s,s',s'')x(s'')ds'ds''
\end{aligned}$$

where

$$R_{\alpha_i,\alpha_j}(s,s',s'') = Z(s,s')^T Q_{ij} Z(s',s'')$$

9.3 Structure

At this point, we introduce data structures and interface with `sosquadvar`. Specifically, using `sosquadvar` with $Z_1 = Z(s, s')$ and $Z_2 = Z(s', s'')$, we get

$$R_{\alpha_i,\alpha_j}(s, s', s'') = Z(s, s')^T Q_{ij} Z(s', s'').$$

For now, we presume this has `dpvar` structure so that

$$R_{\alpha_i,\alpha_j}(s, s', s'') = vec(Q_{ij})^T Z(s, s', s'')$$

$$\begin{aligned}
&(Z^*_{\alpha_i} Q_{ij} Z_{\alpha_j} \mathbf{x})(s) \\
&= vec(Q_{ij})^T \left(\int_{s'} \int_{s''} I_{\alpha_i}(s' - s) I_{\alpha_j}(s' - s'') Z(s, s', s'') x(s'') ds' \right) ds'' \\
&= \int_{s''} vec(Q_{ij})^T \left(\int_{s'} I_{\alpha_i}(s' - s) I_{\alpha_j}(s' - s'') Z(s, s', s'') ds' \right) x(s'') ds''
\end{aligned}$$

re-labeling

$$\begin{aligned}
&(Z^*_{\alpha_i} Q_{ij} Z_{\alpha_j} \mathbf{x})(s) \\
&= \int_{s'} vec(Q_{ij})^T \left(\int_{s''} I_{\alpha_i}(s'' - s) I_{\alpha_j}(s'' - s') Z(s, s'', s') ds'' \right) x(s') ds'
\end{aligned}$$

so that we need only compute

$$\int_{s''} I_{\alpha_i}(s'' - s) I_{\alpha_j}(s'' - s') Z(s, s'', s') ds''$$

However, we obtained an expression for this

$$\int_{s''} I_{\alpha_i}(s'' - s) I_{\alpha_j}(s'' - s') G(s'') ds''$$

in Subsec. 6.1 in terms of mapping $\Delta_\delta Z$. Note, however, that some α_i, α_j pairs get mapped to multiple γ's.

This brings up the question of whether it is better to factor

$$Z(s) \otimes \left(\int_{s''} I_{\alpha_i}(s'' - s) I_{\alpha_j}(s'' - s') Z(s'') ds'' \right) \otimes Z(s')$$

or just do the integration ignoring the s and s' terms. The integration is fairly straightforward, so this may be an option.